

Ziyi Sun

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College of Engineering, Ocean University of China, Qingdao 266100, China

EDUCATION

- **Ocean University of China (985 / Double First-Class)** Sep. 2023 – Jun. 2027 (Expected)
B.Eng. in Mechanical Design, Manufacturing and Automation Qingdao, China
 - GPA: **3.7/4.0** Rank: **1/56** Average: 89.2
 - Selected courses: Advanced Mathematics (97), Linear Algebra (93.5), Probability & Mathematical Statistics (93), Complex Variables & Integral Transforms (94.5), Numerical Methods (96), Mechanical Vibrations (95), Modern Mechanical Design Theory and Methods (100), Underwater Robotics (97.2)

RESEARCH EXPERIENCE

- **Roboparty Lab** Jul. 2026 – Oct. 2026
Frontier Research and Open Source Intern (Mentor: Jagger) Beijing, China
 - Contribute to frontier research and open-source development for embodied intelligence and robot learning systems; related work: [S.1].

PROJECTS

- **Variable-Length Multimodal Biomimetic Robotic Fish for Deep-Sea Inspection** Dec. 2025 – Dec. 2026
Role: Project Leader | Tools: SolidWorks, multimodal vision & sensing OUC SRDP
 - Designed a biomimetic robotic fish with adjustable body length (1.2–1.5 m) and fused visual tracking with water-quality sensing for aquaculture monitoring.
- **Cross-Domain Wire-Driven Biomimetic Eel Robot** Dec. 2024 – Nov. 2025
Role: Project Leader | Tools: vision, Gaussian Splatting, damage detection Provincial | [C.1], [C.2]
 - Built an eel-like robot for offshore wind dynamic-cable inspection with visual tracking, 3D reconstruction, and damage detection; led to IEEE RCAR / ICCCR publications.
- **Streaming Evaluation Framework for Deep-Sea Mining** 2025 – 2026
Role: Lead Contributor | Tools: CAD modeling, multi-AUV simulation Research | [S.2]
 - Built a unified simulation and benchmarking pipeline comparing centralized vs. distributed multi-AUV nodule collection for underwater robot learning.
- **High-Stability Duck-Type Wave Energy Converter with Pendulum PTO** Dec. 2024 – Nov. 2025
Role: Core Member | Tools: AQWA, genetic algorithm, SolidWorks National | [P.1]
 - Improved capture efficiency and output stability via hydrodynamics modeling, shape optimization, and adaptive PTO design (first student inventor on related patent).

PUBLICATIONS & PATENTS

C=CONFERENCE, S=IN SUBMISSION, P=PATENT, SW=SOFTWARE

[C.1] Ziyi Sun, Luwen Li, Changhong He, Shuo Jiang, Zhuoyan Wang, Haoyuan Cheng*.

A Two-Stage Learning Framework for Autonomous Obstacle Avoidance of an Eel-Like Robot.

IEEE RCAR, 2026. **Best Paper Finalist.**

[C.2] Ziyi Sun et al.

Design of Biomimetic Robotic Eel with Omnidirectional Wire-Driven Body and Controlled Central Pattern Generator.

IEEE ICCCR, 2025.

[S.1] Ziyi Sun, Jingwen Chen, Yuxi Wang, Xiuze Xia, Long Cheng, Zhaoxiang Zhang, Junyu Dong.

CHEROE: Every Humanoid Skill as a Traject.

Under review at AAAI 2027 (CCF-A).

[S.2] Ziyi Sun, Changhong He.
A Streaming Evaluation Framework for Deep-Sea Mining: Comparing Centralized and Distributed Nodule Collection.
Under review at *Journal of Marine Science and Engineering (JMSE)*.

[P.1] Ziyi Sun et al.
A High-Stability PTO System for Duck-Type Wave Energy Converters.
Invention Patent (First Student Inventor).

[SW.1] Ziyi Sun.
Underwater Polarized Light Field Prediction Software for Biomimetic Navigation V1.0.
Software Copyright (No. 2024SR0140235).

SKILLS

- **Robot Platform:**Unitree G1
- **Programming & ML:**Python, C/C++, MATLAB, PyTorch, Isaac Gym/Lab, MuJoCo
- **Mechanical Design:**SolidWorks, AutoCAD
- **Research Focus:**Embodied AI, robot learning, sim-to-real transfer, multimodal perception, underwater robotics
- **Languages:**Chinese (Native); English (CET-4: 591, CET-6: 482)

HONORS AND AWARDS

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| • Shandong Provincial Government Scholarship | 2025 |
| • First-Class Comprehensive Scholarship, Ocean University of China | 2024 & 2025 |
| • Outstanding Student / Outstanding Student Cadre, Ocean University of China | 2024 & 2025 |
| • National First Prize, National 3D Digital Innovation Design Competition | 2025 & 2026 |
| • National Second Prize, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM) | 2025 |
| • Honorable Mention, Mathematical Contest in Modeling (MCM) | 2025 |
| • National Third Prize, 18th National College Student Energy Conservation Competition | 2024 |
| • National Gold Award, 3rd “Chuangyi Cup” National Undergraduate Extracurricular Academic Contest | 2024 |
| • National Second Prize, 14th Marine Vehicle Design and Production Competition | 2025 |
| • National First Prize, CRRC Cup National Undergraduate Renewable Energy Competition | 2026 |
| • Science and Technology Innovation Pioneer Team, College of Engineering, OUC | 2024 |

LEADERSHIP EXPERIENCE

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| • Class Representative / Class Monitor, Ocean University of China | 2023 – Present |
| • Team Leader, College of Engineering Debate Team, OUC | 2023 – Present |
| • Director of Event Organization, Student Association for the Study of Socialism with Chinese Characteristics, OUC | 2023 – Present |

VOLUNTEER EXPERIENCE

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| • Volunteer, Qingdao Marathon | 2023 |
| • Member, National College Student Volunteer Outreach Team (Zunyi Conference) | 2024 |
| • Outstanding Individual in Summer Social Practice, Ocean University of China | 2024 |